

Deep Learning Theory

Kernels, infinite width, and effective theory

Notation follows Pehlevan and Bordelon [1] and Roberts, Yaida, and Hanin [2].

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1. Setting

A feedforward MLP of depth L and width N maps $x \in \mathbb{R}^D$ to a scalar by

$$\begin{aligned} h^{(0)} &= x, \\ h^{(\ell+1)} &= N^{-a_{\ell+1}} W^{(\ell+1)} \varphi(h^{(\ell)}), \quad \ell = 0, \dots, L-2, \\ f &= \gamma^{-1} N^{-a_L} w \cdot \varphi(h^{(L-1)}). \end{aligned} \quad (1.1)$$

The activation φ is applied elementwise. Weights are drawn $W_{ij}^{(\ell)} \sim \mathcal{N}(0, N^{-b_\ell})$, and the training set is $\mathcal{D} = \{(x_\mu, y_\mu)\}_{\mu=1}^P$. Two kernels appear throughout:

$$\Phi_{\mu\nu}^{(\ell)} \equiv N^{-1} \varphi(h_\mu^{(\ell)}) \cdot \varphi(h_\nu^{(\ell)}), \quad K_{\mu\nu}(t) \equiv \partial_\theta f(x_\mu; t) \cdot \partial_\theta f(x_\nu; t). \quad (1.2)$$

The first is the *feature kernel* of layer ℓ . The second is the *neural tangent kernel*. In the limits treated here both concentrate, and the $O(N)$ weights are replaced by $O(P^2)$ kernel entries.

The questions are: which functions (1.1) can represent; how $f(\cdot; \theta(t))$ moves under gradient flow; and what the population risk is after training. The answers depend on how (a_ℓ, b_ℓ, γ) scale with N , and on the ratio L/N .

2. Supervised learning

Inputs $x \in \mathcal{X} \subset \mathbb{R}^D$ and labels $y \in \mathcal{Y} \subset \mathbb{R}$ are jointly distributed as $P(x, y)$. A predictor h is taken from a hypothesis class $\mathcal{H}$. For a neural net, $\mathcal{H}$ is the image of $\theta \mapsto f(\cdot; \theta)$.

A loss $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}_{\geq 0}$ compares prediction and label. Two standard choices:

$$\ell_{\text{sq}}(h(x), y) = \frac{1}{2} (h(x) - y)^2, \quad \ell_{01}(h(x), y) = \mathbf{1}[h(x) \neq y]. \quad (2.1)$$

Definition 2.1 (Population and empirical risk).

$$R(h) \equiv \mathbb{E}_{(x,y) \sim P} [\ell(h(x), y)], \quad \hat{R}_{\mathcal{D}}(h) \equiv \frac{1}{P} \sum_{\mu=1}^P \ell(h(x_\mu), y_\mu), \quad (2.2)$$

where $\mathcal{D} = \{(x_\mu, y_\mu)\}_{\mu=1}^P \sim P^{\otimes P}$.

The Bayes predictor $h_{\text{Bayes}} = \arg \min R(h)$ is over all measurable functions and need not lie in $\mathcal{H}$. The best-in-class predictor is $h^* = \arg \min_{h \in \mathcal{H}} R(h)$. Training does not see P . Empirical risk minimization replaces the population problem by

$$h_{\mathcal{D}, \text{ERM}} = \arg \min_{h \in \mathcal{H}} \hat{R}_{\mathcal{D}}(h). \quad (2.3)$$

Lemma 2.1. For any fixed h , $\mathbb{E}_{\mathcal{D}}[\hat{R}_{\mathcal{D}}(h)] = R(h)$.

Proof. Linearity of expectation; the P samples are identically distributed. $\square$

Once h depends on $\mathcal{D}$, the identity fails: $\mathbb{E}_{\mathcal{D}}[\hat{R}_{\mathcal{D}}(h_{\mathcal{D}})]$ is optimistically biased. The generalization gap is

$$\Delta_{\text{gen}}(h_{\mathcal{D}}) \equiv R(h_{\mathcal{D}}) - \hat{R}_{\mathcal{D}}(h_{\mathcal{D}}). \quad (2.4)$$

Generalization error will mean $R(h_{\mathcal{D}})$ itself.

Write $h(x) = f(x; \theta)$. Square loss on $\mathcal{D}$ is

$$\mathcal{L}(\theta) \equiv \frac{1}{2P} \sum_{\mu=1}^P (f(x_\mu; \theta) - y_\mu)^2, \quad (2.5)$$

so $\partial \mathcal{L} / \partial f_\mu = (f_\mu - y_\mu) / P$. Gradient descent and its gradient-flow limit are

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \mathcal{L}(\theta_t), \quad \frac{d\theta}{dt} = -\eta \nabla_{\theta} \mathcal{L}(\theta). \quad (2.6)$$

Exercise 2.1. Verify $\mathbb{E}_{\mathcal{D}}[\hat{R}_{\mathcal{D}}(h)] = R(h)$ from (2.2). Where does the argument use that h does not depend on $\mathcal{D}$?

Solution. Write $\hat{R}_{\mathcal{D}}(h) = P^{-1} \sum_{\mu=1}^P \ell(h(x_\mu), y_\mu)$. Then

$$\mathbb{E}_{\mathcal{D}}[\hat{R}_{\mathcal{D}}(h)] = \frac{1}{P} \sum_{\mu=1}^P \mathbb{E}_{(x_\mu, y_\mu)} [\ell(h(x_\mu), y_\mu)] = \mathbb{E}_{(x, y) \sim P} [\ell(h(x), y)] = R(h),$$

using identical distribution of the samples. The first equality pulls h out of the sample-wise expectation: that is legal only if h is non-random with respect to $\mathcal{D}$. If $h = h_{\mathcal{D}}$, then $\ell(h_{\mathcal{D}}(x_\mu), y_\mu)$ is correlated with the whole sample and the identity fails.

Answer. $\mathbb{E}_{\mathcal{D}}[\hat{R}_{\mathcal{D}}(h)] = R(h)$ for fixed h . The identity uses that h does not depend on $\mathcal{D}$.

3. Feature spaces and kernel machines

3.1 Maximum-margin hyperplanes

A linear classifier $h(x) = \text{sign}(w \cdot x + b)$ on a linearly separable sample admits many separating hyperplanes. The *margin* of (w, b) is the distance to the nearest training point,

$$\delta^* = \min_{\mu \in [P]} \frac{|w \cdot x_\mu + b|}{\|w\|}. \quad (3.1)$$

The scale $(w, b) \sim (\alpha w, \alpha b)$ is redundant. Canonicalize by $\min_{\mu} |w \cdot x_\mu + b| = 1$, so that $\delta^* = 1/\|w\|$. Maximizing the margin is then

$$\min_{w, b} \frac{1}{2} \|w\|^2 \quad \text{subject to} \quad y_\mu (w \cdot x_\mu + b) \geq 1 \quad \forall \mu. \quad (3.2)$$

The Lagrangian is

$$L(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_{\mu=1}^P \alpha_\mu (y_\mu (w \cdot x_\mu + b) - 1), \quad \alpha_\mu \geq 0. \quad (3.3)$$

Stationarity in (w, b) gives the KKT conditions

$$w = \sum_{\mu=1}^P \alpha_\mu y_\mu x_\mu, \quad \sum_{\mu=1}^P \alpha_\mu y_\mu = 0, \quad \alpha_\mu (y_\mu (w \cdot x_\mu + b) - 1) = 0. \quad (3.4)$$

Complementary slackness forces $\alpha_\mu = 0$ unless the constraint is tight. Those points are the *support vectors*; they alone determine w . Substituting (3.4) back into L produces the dual

$$\max_{\alpha \geq 0} \sum_{\mu} \alpha_{\mu} - \frac{1}{2} \sum_{\mu\nu} \alpha_{\mu} \alpha_{\nu} y_{\mu} y_{\nu} x_{\mu} \cdot x_{\nu} \quad \text{s.t.} \quad \sum_{\mu} \alpha_{\mu} y_{\mu} = 0. \quad (3.5)$$

The data enter only through the Gram matrix $G_{\mu\nu} = x_{\mu} \cdot x_{\nu}$. The classifier itself is

$$h(x) = \text{sign} \left(\sum_{\mu} \alpha_{\mu} y_{\mu} x \cdot x_{\mu} + b \right). \quad (3.6)$$

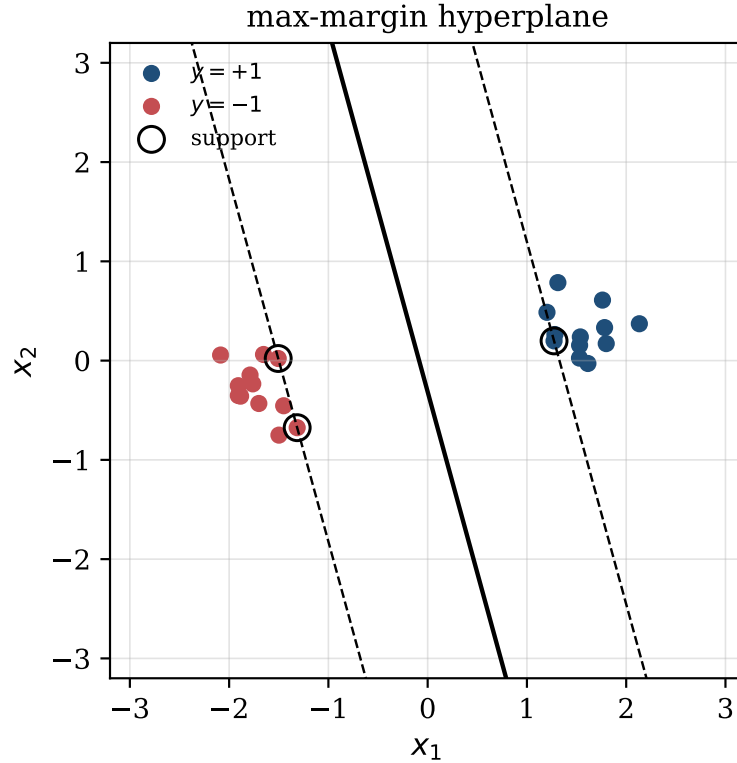


Figure 1: Hard-margin SVM in $\mathbb{R}^2$. Solid line: $w \cdot x + b = 0$. Dashed: $w \cdot x + b = \pm 1$. Circled points are the support vectors.

3.2 The kernel trick

If $\{(x_{\mu}, y_{\mu})\}$ is not linearly separable in $\mathbb{R}^D$, lift through a feature map $\Phi : \mathbb{R}^D \rightarrow \mathbb{R}^M$, $M \geq D$, and run the same SVM on $\Phi(x)$. The dual (3.5) and the predictor (3.6) depend on Φ only through

$$K(x, x') \equiv \Phi(x) \cdot \Phi(x'). \quad (3.7)$$

One never instantiates Φ . It is enough that K be a symmetric positive-definite kernel (Mercer's theorem): there exist $\lambda_k \geq 0$ and orthonormal ψ_k on $L^2(P_x)$ with

$$K(x, x') = \sum_k \lambda_k \psi_k(x) \psi_k(x'). \quad (3.8)$$

The associated feature map is $\Phi(x)_k = \sqrt{\lambda_k} \psi_k(x)$, possibly infinite-dimensional.

Theorem 3.1 (Representer). *Any minimizer of $\hat{R}_{\mathcal{D}}(h) + \lambda \|h\|_{\mathcal{H}_K}^2$ over the RKHS of K is of the form $h(\cdot) = \sum_{\mu=1}^P c_{\mu} K(x_{\mu}, \cdot)$.*

The coefficients $c \in \mathbb{R}^P$ are the only degrees of freedom. Kernel ridge with the square loss (2.5) plus $\frac{\lambda}{2} \|w\|^2$ is derived in §7:

$$f(x) = k(x)^{\top} (\hat{K} + \lambda P I)^{-1} y, \quad \hat{K}_{\mu\nu} = K(x_{\mu}, x_{\nu}), \quad k(x)_{\mu} = K(x, x_{\mu}). \quad (3.9)$$

3.3 A neural kernel

A random one-hidden-layer net

$$f(x) = N^{-1/2} \sum_{i=1}^N v_i \varphi(w_i \cdot x), \quad w_i \sim \mathcal{N}(0, D^{-1} I), \quad v_i \sim \mathcal{N}(0, 1), \quad (3.10)$$

defines, at initialization, the kernel

$$K_{\text{NNGP}}(x, x') = \mathbb{E}_{w,v} [f(x) f(x')] = \mathbb{E}_w [\varphi(w \cdot x) \varphi(w \cdot x')]. \quad (3.11)$$

For $\varphi(z) = \max(z, 0)$ this is the Cho–Saul kernel. Derive it as follows. Let $w \sim \mathcal{N}(0, I)$, set $u = \hat{x} \cdot w$ and $u' = \hat{x}' \cdot w$ with $\hat{x} = x/\|x\|$, and write $\vartheta = \arccos(\hat{x} \cdot \hat{x}')$. Then (u, u') is a standard bivariate Gaussian with correlation $\cos \vartheta$, equivalently the two coordinates of a 2d standard Gaussian g in the plane of $(\hat{x}, \hat{x}')$:

$$u = \rho \cos \phi, \quad u' = \rho \cos(\phi - \vartheta), \quad g = (\rho \cos \phi, \rho \sin \phi). \quad (3.12)$$

ReLU is supported on $u > 0$, $u' > 0$, a wedge of opening $\pi - \vartheta$, namely $\phi \in (\vartheta - \pi/2, \pi/2)$. The Gaussian measure on the plane is $e^{-\rho^2/2} \rho d\rho d\phi / (2\pi)$, so

$$\mathbb{E}[u_+ u'_+] = \int_{\vartheta - \pi/2}^{\pi/2} d\phi \int_0^{\infty} d\rho \frac{\rho^3 e^{-\rho^2/2}}{2\pi} \cos \phi \cos(\phi - \vartheta). \quad (3.13)$$

The radial integral is $\int_0^{\infty} \rho^3 e^{-\rho^2/2} d\rho = 2$. For the angle, $\cos \phi \cos(\phi - \vartheta) = \frac{1}{2} (\cos(2\phi - \vartheta) + \cos \vartheta)$, so

$$\int_{\vartheta - \pi/2}^{\pi/2} \cos \phi \cos(\phi - \vartheta) d\phi = \left[\frac{1}{4} \sin(2\phi - \vartheta) + \frac{\phi}{2} \cos \vartheta \right]_{\vartheta - \pi/2}^{\pi/2} = \frac{1}{2} (\sin \vartheta + (\pi - \vartheta) \cos \vartheta). \quad (3.14)$$

Hence $\mathbb{E}[u_+ u'_+] = (2\pi)^{-1} (\sin \vartheta + (\pi - \vartheta) \cos \vartheta)$. Scaling back $w \sim \mathcal{N}(0, D^{-1} I)$ multiplies by $\|x\| \|x'\| / D$:

$$\mathbb{E}[\varphi(w \cdot x) \varphi(w \cdot x')] = \frac{\|x\| \|x'\|}{2\pi D} (\sin \vartheta + (\pi - \vartheta) \cos \vartheta). \quad (3.15)$$

Deeper random nets compose this expectation layer by layer: that is the NNGP recursion below.

Exercise 3.1. From (3.4), show that if x_{μ} is not a support vector then $\alpha_{\mu} = 0$. Conclude that w is a linear combination of the support vectors only.

Solution. Complementary slackness in (3.4) is $\alpha_{\mu} (y_{\mu} (w \cdot x_{\mu} + b) - 1) = 0$. If x_{μ} is not a support vector the inequality is slack, $y_{\mu} (w \cdot x_{\mu} + b) > 1$, so the second factor is nonzero and therefore $\alpha_{\mu} = 0$. The stationarity condition $w = \sum_{\nu} \alpha_{\nu} y_{\nu} x_{\nu}$ then retains only those ν with $\alpha_{\nu} \neq 0$, i.e. the support vectors.

Answer. x_{μ} not on the margin $\Rightarrow \alpha_{\mu} = 0$, hence $w = \sum_{\text{SV}} \alpha_{\nu} y_{\nu} x_{\nu}$.

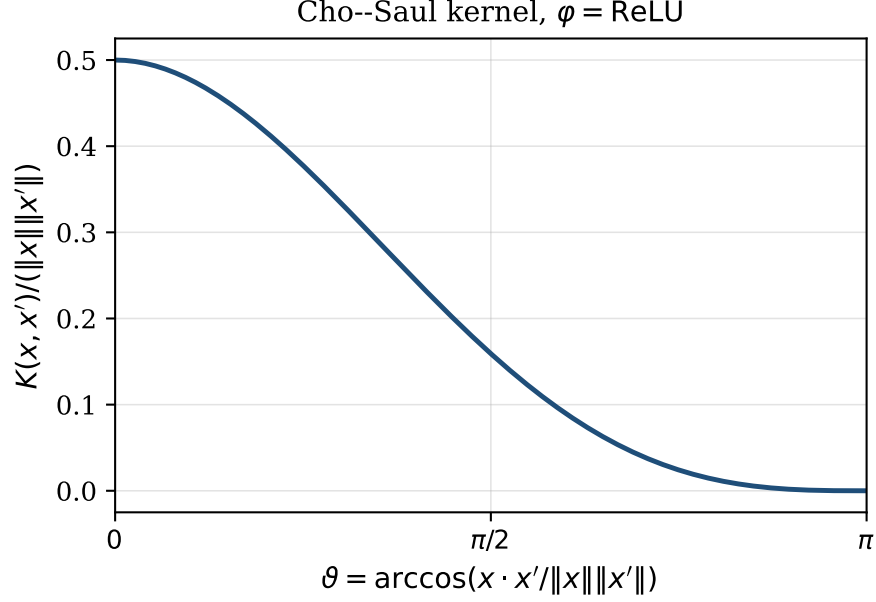


Figure 2: Unit-variance Cho–Saul factor $\mathbb{E}[u_+ u'_+] = (2\pi)^{-1}(\sin \vartheta + (\pi - \vartheta) \cos \vartheta)$ as a function of the angle between x and x' .

4. Training dynamics and the NTK

Let $f(x; \theta)$ be scalar, $f_\mu \equiv f(x_\mu; \theta)$, and $\Delta_\mu \equiv -\partial \ell_\mu / \partial f_\mu$. For (2.5),

$$\Delta_\mu = y_\mu - f_\mu. \quad (4.1)$$

4.1 Function-space gradient flow

Chain rule on (2.6):

$$\frac{df(x)}{dt} = \partial_\theta f(x) \cdot \frac{d\theta}{dt} = -\eta \partial_\theta f(x) \cdot \nabla_\theta \mathcal{L}. \quad (4.2)$$

The loss gradient is $\nabla_\theta \mathcal{L} = P^{-1} \sum_\mu (\partial \ell_\mu / \partial f_\mu) \partial_\theta f_\mu$, so

$$\partial_t f(x) = \frac{\eta}{P} \sum_{\mu=1}^P \underbrace{(\partial_\theta f(x) \cdot \partial_\theta f(x_\mu))}_{K(x, x_\mu; t)} \Delta_\mu. \quad (4.3)$$

Definition 4.1 (Neural tangent kernel).

$$K(x, x'; t) \equiv \partial_\theta f(x; \theta(t)) \cdot \partial_\theta f(x'; \theta(t)). \quad (4.4)$$

On the training set, $\hat{K}_{\mu\nu}(t) \equiv K(x_\mu, x_\nu; t)$ and $k(x; t)_\mu \equiv K(x, x_\mu; t)$.

K is a Gram matrix of Jacobians, hence positive semidefinite. Equation (4.3) is exact at any finite width. It is not closed: $K(t)$ depends on $\theta(t)$.

4.2 Frozen kernel

Suppose $K(x, x'; t) = K(x, x'; 0) \equiv K(x, x')$. On the training points,

$$\partial_t F = \frac{\eta}{P} \hat{K} (y - F), \quad F_\mu(t) \equiv f(x_\mu; t). \quad (4.5)$$

Diagonalize $\hat{K} = U \Lambda U^\top$, $\Lambda = \text{diag}(\lambda_i)$. In the eigenbasis $F' = U^\top F$, $y' = U^\top y$,

$$\partial_t F'_i = \frac{\eta \lambda_i}{P} (y'_i - F'_i) \implies F'_i(t) = y'_i + e^{-(\eta \lambda_i / P)t} (F'_i(0) - y'_i). \quad (4.6)$$

Mode i trains on the timescale $P/(\eta \lambda_i)$.

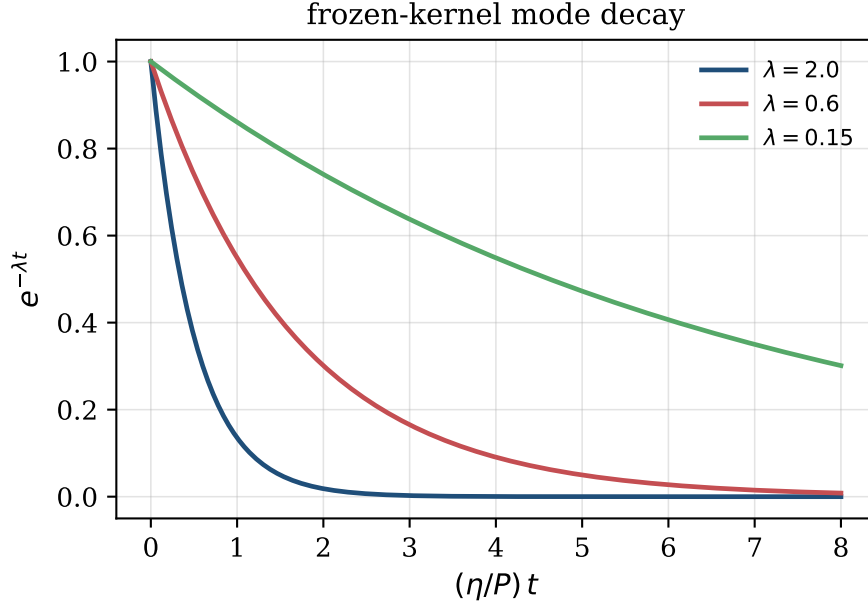


Figure 3: Eigenmodes of a frozen kernel. Residual of mode i decays as $e^{-(\eta \lambda_i / P)t}$.

Transforming back,

$$F(t) = y + e^{-(\eta/P)\hat{K}t} (F(0) - y) = F(0) + (I - e^{-(\eta/P)\hat{K}t}) (y - F(0)). \quad (4.7)$$

A test point is slaved to the training residual. Substituting (4.7) into (4.3),

$$f(x; t) = f(x; 0) + k(x)^\top \hat{K}^{-1} (I - e^{-(\eta/P)\hat{K}t}) (y - F(0)), \quad (4.8)$$

with the pseudoinverse if $\hat{K}$ is singular (the component of $y - F(0)$ in $\ker \hat{K}$ never trains). As $t \rightarrow \infty$ with $f(\cdot; 0) = 0$,

$$f(x; \infty) = k(x)^\top \hat{K}^{-1} y, \quad (4.9)$$

which is (3.9) at $\lambda = 0$.

4.3 Linear models

If $f(x; w) = w \cdot \Phi(x)$ with Φ fixed, then

$$\frac{dw}{dt} = \frac{\eta}{P} \sum_{\mu} (y_{\mu} - w \cdot \Phi(x_{\mu})) \Phi(x_{\mu}), \quad (4.10)$$

and

$$\partial_t f(x) = \frac{\eta}{P} \sum_{\mu} K_{\Phi}(x, x_{\mu})(y_{\mu} - f(x_{\mu})), \quad K_{\Phi}(x, x') = \Phi(x) \cdot \Phi(x'). \quad (4.11)$$

The NTK of a linear model is K_{Φ} , and it cannot move. Adding $\frac{\lambda}{2}\|w\|^2$ to (2.5) replaces $\hat{K}^{-1}$ by $(\hat{K} + \lambda PI)^{-1}$; see §7.

4.4 Two-layer NTK

For $f(x) = N^{-1/2}v \cdot \varphi(Wx/\sqrt{D})$ the Jacobian splits over v and W :

$$\begin{aligned} \partial_{v_i} f(x) &= N^{-1/2} \varphi(h_i(x)), \\ \partial_{W_{ij}} f(x) &= N^{-1/2} D^{-1/2} v_i \dot{\varphi}(h_i(x)) x_j, \end{aligned} \quad (4.12)$$

with $h(x) = Wx/\sqrt{D}$. Therefore

$$K(x, x') = N^{-1} \varphi(h(x)) \cdot \varphi(h(x')) + N^{-1} D^{-1} (v \odot \dot{\varphi}(h(x))) \cdot (v \odot \dot{\varphi}(h(x'))) (x \cdot x'). \quad (4.13)$$

The first term is the feature kernel Φ . The second is the contribution of the first-layer weights. At infinite N both concentrate, and if v, W move by $o_N(1)$ then $K(t) = K(0)$.

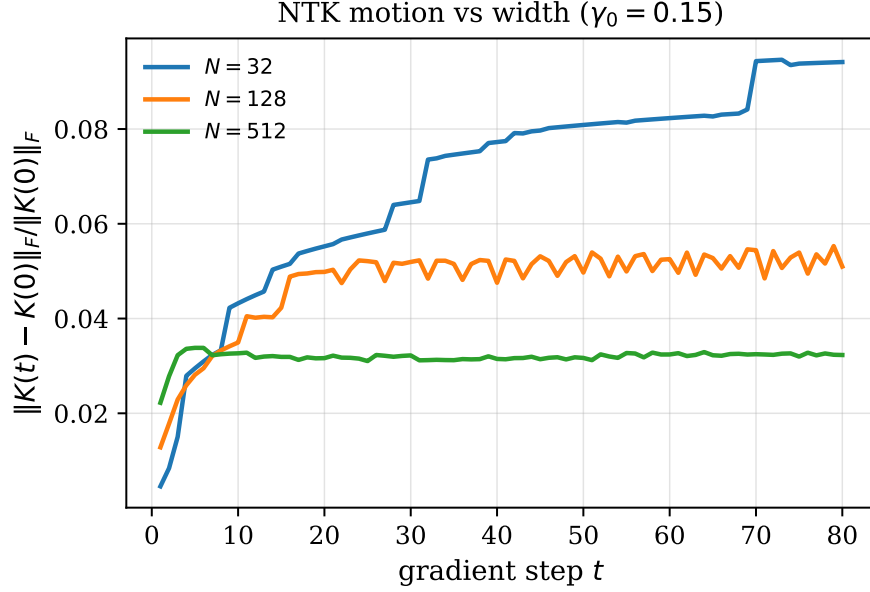


Figure 4: Two-layer ReLU, $\gamma_0 = 0.15$, square loss. Relative Frobenius change of the empirical NTK on the training set, for widths $N = 32, 128, 512$.

4.5 Expressivity

Theorem 4.1 (Cybenko, Hornik). *If φ is continuous, bounded, and nonconstant, then shallow nets $x \mapsto \sum_{i=1}^N v_i \varphi(w_i \cdot x + b_i)$ are dense in $C([0, 1]^D)$ in the uniform norm.*

Density does not control the number of neurons. For ReLU nets of depth L and width N on N_0 inputs, each hidden unit folds the input space along a hyperplane. A lower bound on the number

of linear regions is

$$\Omega\left(\left(\frac{N}{N_0}\right)^{(L-1)N_0} N^{N_0}\right). \quad (4.14)$$

A matched-parameter shallow net ($L = 1$, width NL) has only $O((NL)^{N_0})$ regions. Depth buys exponentially many linear pieces per parameter.

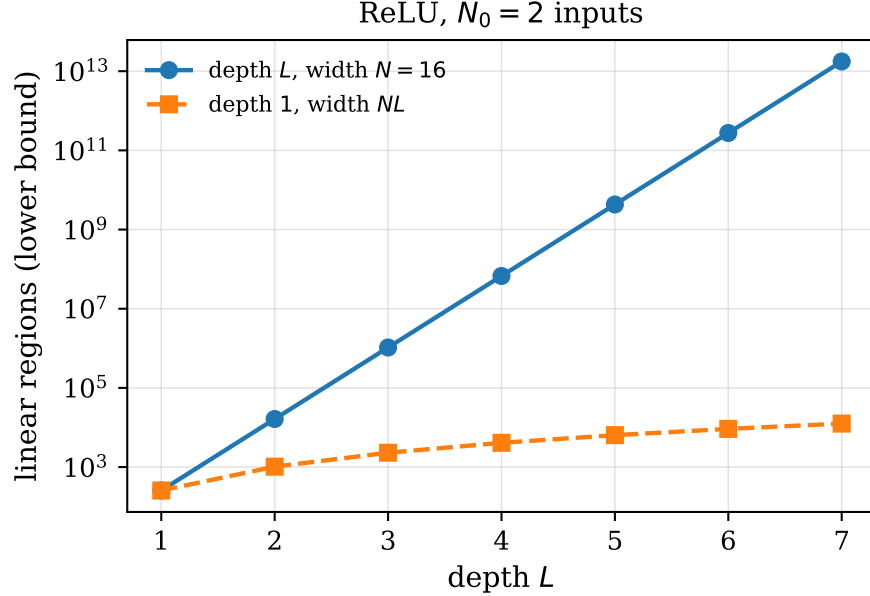


Figure 5: Lower bound (4.14) versus a shallow net with the same number of units NL , for $N_0 = 2$, $N = 16$.

Exercise 4.1. Starting from (4.5), derive (4.7) by expanding F in the eigenbasis of $\hat{K}$. What is the training time of a mode with kernel eigenvalue λ ?

Solution. Let $\hat{K} = U\Lambda U^\top$ with $\Lambda = \text{diag}(\lambda_i)$ and set $F' = U^\top F$, $y' = U^\top y$. Substitute into (4.5):

$$\partial_t F' = \frac{\eta}{P} \Lambda (y' - F'),$$

which decouples: $\partial_t F'_i = (\eta\lambda_i/P)(y'_i - F'_i)$. The ODE $\dot{u} = -a(u - b)$ with $a = \eta\lambda_i/P$ and $b = y'_i$ integrates to

$$F'_i(t) = y'_i + e^{-(\eta\lambda_i/P)t} (F'_i(0) - y'_i).$$

In matrix form $F'(t) = y' + e^{-(\eta/P)\Lambda t} (F'(0) - y')$. Multiplying by U and using $Ue^{-(\eta/P)\Lambda t}U^\top = e^{-(\eta/P)\hat{K}t}$ gives (4.7). Mode i has time constant $P/(\eta\lambda_i)$.

Answer. $F(t) = y + e^{-(\eta/P)\hat{K}t} (F(0) - y)$. Mode with eigenvalue λ trains on the timescale $P/(\eta\lambda)$.

5. Lazy versus rich parameterization

Whether $K(t)$ stays at $K(0)$ is a scaling question [1]. Write

$$h_\mu^{(1)} = N^{-a_1} W^{(1)} x_\mu, \quad h_\mu^{(\ell+1)} = N^{-a_{\ell+1}} W^{(\ell+1)} \varphi(h_\mu^{(\ell)}), \quad f_\mu = \gamma^{-1} N^{-a_L} w \cdot \varphi(h_\mu^{(L-1)}), \quad (5.1)$$

with $W_{ij}^{(\ell)} \sim \mathcal{N}(0, N^{-b_\ell})$ and learning rate

$$\eta = \eta_0 \gamma^2 N^c, \quad \gamma = \gamma_0 N^d, \quad \gamma_0 = O_N(1). \quad (5.2)$$

Backpropagated fields:

$$g_{\mu,i}^{(\ell)} \equiv N^{a_\ell + \frac{1}{2}b_\ell} \frac{\partial f_\mu}{\partial h_{\mu,i}^{(\ell)}}, \quad G_{\mu\nu}^{(\ell)} \equiv N^{-1} g_\mu^{(\ell)} \cdot g_\nu^{(\ell)}. \quad (5.3)$$

Preactivations $O_N(1)$ at initialization. For $\ell \geq 2$,

$$\mathbb{E}[|h_{\mu i}^{(\ell)}|^2] = N^{-2a_\ell - b_\ell} \mathbb{E}[|\varphi(h_\mu^{(\ell-1)})|^2]. \quad (5.4)$$

A finite nonzero limit requires

$$2a_\ell + b_\ell = 1, \quad \ell = 2, \dots, L, \quad 2a_1 + b_1 = 0. \quad (5.5)$$

Under (5.5) the hidden units of a layer become i.i.d. Gaussians and $\Phi^{(\ell)}$ concentrates on

$$\Phi_{\mu\nu}^{(\ell)} = \mathbb{E}_{h \sim \mathcal{N}(0, \Phi^{(\ell-1)})} [\varphi(h_\mu) \varphi(h_\nu)], \quad \Phi_{\mu\nu}^{(0)} = D^{-1} x_\mu \cdot x_\nu. \quad (5.6)$$

Predictions move in $O_N(1)$ time. From (4.3), $\partial_t f = O_N(1)$ requires $\eta K/P = O_N(1)$. Expanding K over layers as in (4.13) and using $\Phi, G = O_N(1)$,

$$\begin{aligned} \frac{\gamma^2}{N^c} \partial_\theta f_\mu \cdot \partial_\theta f_\nu &= N^{-c-2a_L+1} \Phi_{\mu\nu}^{(L-1)} + \sum_{\ell=2}^{L-1} N^{-c-2a_\ell+1} G_{\mu\nu}^{(\ell)} \Phi_{\mu\nu}^{(\ell-1)} \\ &\quad + N^{-c-2a_1} G_{\mu\nu}^{(1)} \Phi_{\mu\nu}^{(0)}. \end{aligned} \quad (5.7)$$

Every exponent vanishes iff

$$2a_\ell + c = 1 \quad (\ell \geq 2), \quad 2a_1 + c = 0. \quad (5.8)$$

Features move, or they do not. Gradient flow on $W^{(1)}$ gives

$$\partial_t h_\mu^{(1)} = N^{-2a_1-c+d-\frac{1}{2}} \frac{\eta_0 \gamma_0}{P} \sum_\nu \Delta_\nu g_\nu^{(1)} \Phi_{\mu\nu}^{(0)}. \quad (5.9)$$

Imposing $\partial_t h^{(1)} = O_N(1)$ together with (5.8) forces $d = \frac{1}{2}$. The same exponent appears at every later layer, because

$$\partial_t h_\mu^{(\ell)} = N^{-2a_\ell-c+d-\frac{1}{2}} \times (O_N(1)) + N^{-a_\ell} W^{(\ell)}(\dot{\varphi}(h_\mu^{(\ell-1)}) \odot \partial_t h_\mu^{(\ell-1)}) \quad (5.10)$$

and the second term inherits the scaling of $\partial_t h^{(\ell-1)}$. Any $d < \frac{1}{2}$ sends $\partial_t h \rightarrow 0$: features freeze and $K(t) \rightarrow K(0)$. The NTK parameterization of Jacot et al. is $d = 0$.

Proposition 5.1. *The family $d = \frac{1}{2}$, (5.5), (5.8) yields infinite-width feature learning. A representative (mean-field / μP) is*

$$a_1 = 0, \quad a_{\ell \geq 2} = \frac{1}{2}, \quad b_\ell = 0, \quad c = 0, \quad d = \frac{1}{2}, \quad \eta = \eta_0 \gamma_0^2 N, \quad f = \frac{w \cdot \varphi(h^{(L-1)})}{\gamma_0 \sqrt{N}}. \quad (5.11)$$

Then $f_\mu(0) = O_N(N^{-1/2})$. The NTK parameterization is the same architecture with $d = 0$, or equivalently $\gamma_0 \rightarrow 0$ after a time rescaling.

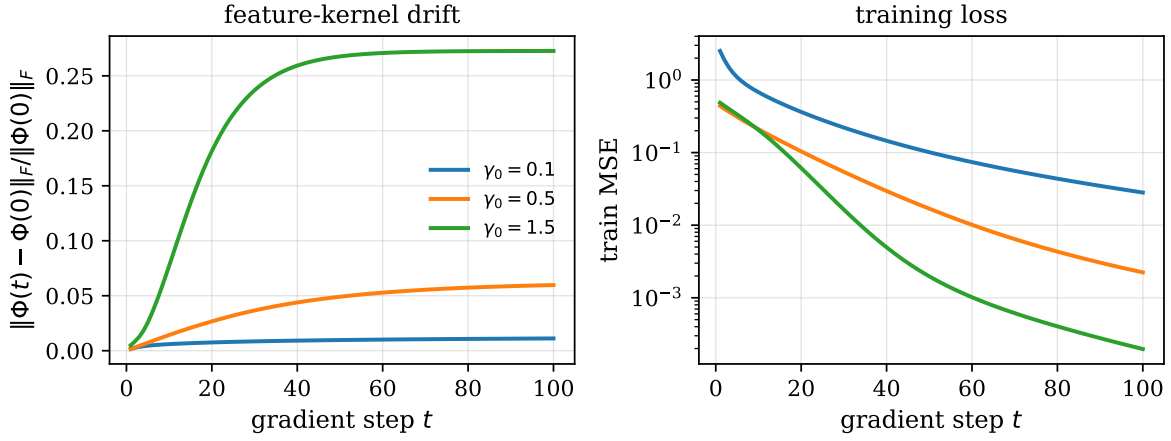
lazy vs rich, $N = 256$ 

Figure 6: Two-layer ReLU, $N = 256$, $\eta = \eta_0 \gamma_0^2 N$. Left: relative motion of the hidden feature kernel Φ . Right: training square loss. Only γ_0 is varied.

Chizat's diagnostic: training $\alpha f(\theta/\alpha)$ as $\alpha \rightarrow \infty$ linearizes about initialization. Here $\alpha \sim \gamma_0^{-1} N^{1/2-d}$.

Exercise 5.1. Take $\gamma = \gamma_0 N^d$, $\eta = \eta_0 \gamma_0^2 N^{2d}$ (i.e. $c = 0$), and

$$f = (\gamma_0 N^d \sqrt{N})^{-1} v \cdot \varphi(Wx/\sqrt{D}),$$

with $v_i, W_{ij} = O_N(1)$. Compute the N -scaling of $\partial_t W$ and $\partial_t v$. Recover $d = \frac{1}{2}$ as the condition that W moves by $O_N(1)$.

Solution. Write $s = (\gamma_0 N^d \sqrt{N})^{-1}$ and $h = Wx/\sqrt{D}$, so $f = s v \cdot \varphi(h)$. Jacobians:

$$\partial_{v_i} f = s \varphi(h_i), \quad \partial_{W_{ij}} f = s v_i \dot{\varphi}(h_i) x_j / \sqrt{D}.$$

Gradient flow on (2.5) is $\partial_t \theta = -\eta P^{-1} \sum_{\mu} (f_{\mu} - y_{\mu}) \partial_{\theta} f_{\mu}$, i.e. $\partial_t \theta = (\eta/P) \sum_{\mu} \Delta_{\mu} \partial_{\theta} f_{\mu}$. The prefactor is

$$\frac{\eta s}{P} = \frac{\eta_0 \gamma_0^2 N^{2d}}{P} \cdot \frac{1}{\gamma_0 N^d \sqrt{N}} = \frac{\eta_0 \gamma_0}{P} N^{d-1/2}.$$

With $\Delta, \varphi, v, x = O_N(1)$,

$$\partial_t v_i = O_N(N^{d-1/2}), \quad \partial_t W_{ij} = O_N(N^{d-1/2}).$$

An $O_N(1)$ change in W over $O(1)$ time requires $d - \frac{1}{2} = 0$, i.e. $d = \frac{1}{2}$. For $d = 0$ both increments are $O_N(N^{-1/2})$ and the hidden features freeze.

Answer. $\partial_t v, \partial_t W = O_N(N^{d-1/2})$. Hidden weights move by $O_N(1)$ iff $d = \frac{1}{2}$.

6. Infinite-width description

6.1 NNGP recursion

Fix (5.11) at initialization. A single first-layer unit is

$$h_i^{(1)}(x) = D^{-1/2} \sum_{j=1}^D W_{ij}^{(1)} x_j, \quad W_{ij}^{(1)} \sim \mathcal{N}(0, 1). \quad (6.1)$$

The pair $(h_i^{(1)}(x), h_i^{(1)}(x'))$ is Gaussian with covariance $\Phi^{(0)}(x, x') = D^{-1} x \cdot x'$, and distinct neurons are independent.

Layer $\ell + 1$ is a sum of N i.i.d. terms. Because $W^{(\ell+1)}$ is Gaussian,

$$h_i^{(\ell+1)}(\cdot) \xrightarrow{N \rightarrow \infty} \text{GP}(0, \Phi^{(\ell)}), \quad \Phi^{(\ell)}(x, x') = \mathbb{E}[\varphi(h^{(\ell)}(x)) \varphi(h^{(\ell)}(x'))], \quad (6.2)$$

the expectation being over the GP of layer ℓ [1]. The network output is then a GP with kernel $\Phi^{(L-1)}$. For two inputs the expectation is a two-dimensional Gaussian integral against φ .

6.2 NTK recursion

Write $f^{(\ell+1)} = N^{-1/2} W^{(\ell+1)} \varphi(h^{(\ell)})$ and $\dot{\varphi} \equiv \varphi'$. Differentiating through a layer,

$$\partial_\theta f^{(\ell+1)}(x) = N^{-1/2} (\partial_\theta W^{(\ell+1)}) \varphi(h^{(\ell)}(x)) + N^{-1/2} W^{(\ell+1)} (\dot{\varphi}(h^{(\ell)}(x)) \odot \partial_\theta h^{(\ell)}(x)). \quad (6.3)$$

The inner product of two such Jacobians splits into a new-weights term and a previous-layer term. After averaging over $W^{(\ell+1)} \sim \mathcal{N}(0, 1)$ and sending $N \rightarrow \infty$,

$$\begin{aligned} \Phi^{(0)}(x, x') &= D^{-1} x \cdot x', \\ \Phi^{(\ell+1)}(x, x') &= \mathbb{E}[\varphi(h) \varphi(h')], \\ \dot{\Phi}^{(\ell+1)}(x, x') &= \mathbb{E}[\dot{\varphi}(h) \dot{\varphi}(h')], \\ \Theta^{(1)}(x, x') &= \Phi^{(0)}(x, x'), \\ \Theta^{(\ell+1)}(x, x') &= \Phi^{(\ell)}(x, x') + \dot{\Phi}^{(\ell)}(x, x') \Theta^{(\ell)}(x, x'), \end{aligned} \quad (6.4)$$

where (h, h') is the NNGP pair at layer ℓ [1]. The first term in the last line is the kernel of the incoming activations; the second gates the previous NTK by $\dot{\varphi}$. In NTK scaling ($d = 0$), $\Theta(t) = \Theta(0)$ and (4.8) applies with $K = \Theta^{(L)}$.

6.3 Two-layer DMFT

Return to (5.11) with $L = 2$:

$$h_\mu(t) = D^{-1/2} W(t) x_\mu, \quad f_\mu(t) = \gamma_0^{-1} N^{-1/2} v(t) \cdot \varphi(h_\mu(t)). \quad (6.5)$$

Integrating the gradient flow,

$$v(t) = v(0) + \frac{\eta_0 \gamma_0}{P \sqrt{N}} \int_0^t ds \sum_\nu \Delta_\nu(s) \varphi(h_\nu(s)), \quad (6.6)$$

$$W(t) = W(0) + \frac{\eta_0 \gamma_0}{P \sqrt{D}} \int_0^t ds \sum_\nu \Delta_\nu(s) g_\nu(s) x_\nu^\top, \quad (6.7)$$

with $g_{\nu,i} \propto v_i \dot{\varphi}(h_{\nu i})$. Substitute into h_μ :

$$h_\mu(t) = \underbrace{D^{-1/2}W(0)x_\mu}_{\chi_\mu(t)} + \frac{\eta_0\gamma_0}{P} \int_0^t ds \sum_\nu \Delta_\nu(s) G_{\mu\nu}(t, s) \Phi_{\mu\nu}^{(0)}, \quad (6.8)$$

up to the precise index placement on $\Phi^{(0)}$. The initial field χ is Gaussian with covariance $\Phi^{(0)}$. The kernels

$$\Phi_{\mu\nu}(t, s) = N^{-1} \varphi(h_\mu(t)) \cdot \varphi(h_\nu(s)), \quad G_{\mu\nu}(t, s) = N^{-1} g_\mu(t) \cdot g_\nu(s) \quad (6.9)$$

concentrate as $N \rightarrow \infty$ on deterministic two-time order parameters [1]. Self-consistency: Φ and G computed from the process (6.8) must reproduce (6.9).

The integral term in (6.8) is proportional to γ_0 . Sending $\gamma_0 \rightarrow 0$ freezes $h(t) = h(0)$ and recovers

$$\partial_t f_\mu = \frac{\eta_0}{P} \sum_\nu \Theta_{\mu\nu}(0) \Delta_\nu. \quad (6.10)$$

For $L > 2$ each layer has its own pair $(\Phi^{(\ell)}, G^{(\ell)})$, and the preactivations are coupled GPs across time. The closure is still $O(P^2)$ kernels rather than $O(N)$ neurons.

If $\varphi(h) = h$, the kernels stay Gaussian and the dynamics reduce to an SVD flow: input–output modes train independently, with time constants set by their singular values [1]. Infinite width does not by itself freeze features; a linear network in μP scaling still moves its internal SVD.

Exercise 6.1. From (6.6)–(6.7), show that $\gamma_0 \rightarrow 0$ at fixed η_0 freezes W while still allowing v to interpolate on an $O(1)$ timescale.

Solution. The increment of the first-layer weights is

$$W(t) - W(0) = \frac{\eta_0\gamma_0}{P\sqrt{D}} \int_0^t ds \sum_\nu \Delta_\nu(s) g_\nu(s) x_\nu^\top = O(\gamma_0).$$

Thus $W(t) \rightarrow W(0)$ as $\gamma_0 \rightarrow 0$, and $h_\mu(t)$ freezes. The readout increment is

$$v(t) - v(0) = \frac{\eta_0\gamma_0}{P\sqrt{N}} \int_0^t ds \sum_\nu \Delta_\nu(s) \varphi(h_\nu(s)) = O(\gamma_0 N^{-1/2}).$$

This also vanishes, but $f = \gamma_0^{-1} N^{-1/2} v \cdot \varphi(h)$ compensates with $1/\gamma_0$. The change in f coming from Δv alone is

$$\begin{aligned} \Delta f_\mu &= \gamma_0^{-1} N^{-1/2} (v(t) - v(0)) \cdot \varphi(h_\mu) \\ &= \frac{\eta_0}{P} N^{-1} \int_0^t ds \sum_\nu \Delta_\nu(s) \varphi(h_\mu) \cdot \varphi(h_\nu), \end{aligned}$$

which is $O(1)$ and independent of γ_0 . In the limit one recovers the frozen-NTK ODE (6.10), whose solution (4.7) interpolates on the timescale $P/(\eta_0\lambda_{\min})$.

Answer. W freezes as $O(\gamma_0)$. The $1/\gamma_0$ in f cancels the γ_0 in Δv , so the readout still interpolates in $O(1)$ time.

7. Generalization in the lazy regime

Once K is frozen, training is kernel ridge (3.9). The three views below are the same estimator.

7.1 Primal, dual, and push-through

Let $\Phi(x) \in \mathbb{R}^M$ be a fixed feature map, $X \in \mathbb{R}^{P \times M}$ the design with rows $\Phi(x_\mu)^\top$, and $y \in \mathbb{R}^P$. Square loss (2.5) plus weight decay is

$$\mathcal{L}(w) = \frac{1}{2P} \|Xw - y\|^2 + \frac{\lambda}{2} \|w\|^2. \quad (7.1)$$

Stationarity $\nabla_w \mathcal{L} = 0$ reads

$$\left(\frac{1}{P} X^\top X + \lambda I\right)w = \frac{1}{P} X^\top y \quad \Longleftrightarrow \quad (X^\top X + \lambda P I)w = X^\top y. \quad (7.2)$$

Hence the primal estimator

$$\hat{w} = \frac{1}{P} \left(\frac{1}{P} X^\top X + \lambda I\right)^{-1} X^\top y. \quad (7.3)$$

Push-through. The two matrices $X^\top X + \lambda P I$ and $XX^\top + \lambda P I$ satisfy

$$(X^\top X + \lambda P I)X^\top = X^\top (XX^\top + \lambda P I), \quad (7.4)$$

because both sides equal $X^\top XX^\top + \lambda PX^\top$. Left-multiply by $(X^\top X + \lambda P I)^{-1}$ and right-multiply by $(XX^\top + \lambda P I)^{-1}$:

$$X^\top (XX^\top + \lambda P I)^{-1} = (X^\top X + \lambda P I)^{-1} X^\top. \quad (7.5)$$

Substitute into (7.2):

$$\hat{w} = X^\top (\hat{K} + \lambda P I)^{-1} y, \quad \hat{K} \equiv XX^\top. \quad (7.6)$$

On a test point, $f(x) = \Phi(x) \cdot \hat{w}$ is (3.9). If $P > M$ and $\lambda \downarrow 0$, (7.3) is ordinary least squares. If $P < M$ and $\lambda \downarrow 0$, (7.6) is the min-norm interpolant $\hat{w} = X^\top \hat{K}^{-1} y$.

Dynamics. Gradient flow on (7.1) starting at $w = 0$ is

$$\partial_t w = -\eta \left(\frac{1}{P} X^\top (Xw - y) + \lambda w \right). \quad (7.7)$$

As $t \rightarrow \infty$ one recovers (7.2). At $\lambda = 0$ this is (4.10), and the function-space solution is (4.9).

7.2 Bayes: completing the square

Likelihood $y \mid X, w \sim \mathcal{N}(Xw, \sigma^2 I)$ and prior $w \sim \mathcal{N}(0, \sigma_0^2 I)$ have joint exponent

$$-\frac{\|Xw - y\|^2}{2\sigma^2} - \frac{\|w\|^2}{2\sigma_0^2} = -\frac{1}{2\sigma^2} \left(w^\top (X^\top X + \frac{\sigma^2}{\sigma_0^2} I) w - 2w^\top X^\top y + |y|^2 \right). \quad (7.8)$$

Completing the square, the posterior is Gaussian with mean

$$w_{\text{MAP}} = (X^\top X + \frac{\sigma^2}{\sigma_0^2} I)^{-1} X^\top y. \quad (7.9)$$

Match to (7.2): $\lambda P = \sigma^2 / \sigma_0^2$, i.e.

$$\lambda = \frac{\sigma^2}{P\sigma_0^2}. \quad (7.10)$$

The ridge parameter in (7.1) is the prior precision in units of the $1/(2P)$ loss.

7.3 Population eigenbasis

On $L^2(P_x)$ the kernel has Mercer eigenvalues λ_k (not the ridge parameter) and eigenfunctions ψ_k ,

$$\int K(x, x') \psi_k(x') dP_x(x') = \lambda_k \psi_k(x), \quad \lambda_1 \geq \lambda_2 \geq \dots \geq 0. \quad (7.11)$$

Expand the target $f^*(x) = \mathbb{E}[y \mid x] = \sum_k \bar{a}_k \psi_k$, and write $t_k \equiv \bar{a}_k^2$. Additive label noise has variance σ^2 . Features are $\Phi(x)_k = \sqrt{\lambda_k} \psi_k(x)$.

7.4 Effective ridge

Write $\Sigma = \mathbb{E}_x[\Phi(x)\Phi(x)^\top] = \text{diag}(\lambda_k)$ in the Mercer basis, and $\hat{\Sigma} = P^{-1}X^\top X$. Let $\hat{\Sigma}^{-\mu}$ be the empirical covariance without sample μ . Sherman–Morrison on (7.2) gives the leave-one-out relation

$$\Phi(x_\mu) \cdot \hat{w} = \frac{\Phi(x_\mu) \cdot \hat{w}^{-\mu}}{1 + P^{-1}\Phi(x_\mu)^\top (\hat{\Sigma}^{-\mu} + \lambda I)^{-1} \Phi(x_\mu)}. \quad (7.12)$$

The denominator is $1 + h_{\mu\mu}$, with leverage $h_{\mu\mu}$. In the proportional limit every leverage concentrates on the same value n_{eff}/P , and $(\hat{\Sigma}^{-\mu} + \lambda I)^{-1}$ concentrates on a deterministic matrix $(\Sigma + \kappa I)^{-1}$ sharing the Mercer eigenbasis. Then

$$n_{\text{eff}} = \mathbb{E}[\Phi(x)^\top (\Sigma + \kappa I)^{-1} \Phi(x)] = \sum_k \frac{\lambda_k}{\lambda_k + \kappa}. \quad (7.13)$$

Self-consistency of the equivalent ridge—the in-sample problem is a ridge- λ estimator after n_{eff} degrees of freedom have been spent—is $\kappa(1 - n_{\text{eff}}/P) = \lambda$, or

$$\kappa = \frac{\lambda}{1 - n_{\text{eff}}/P}. \quad (7.14)$$

(The same κ appears as the deterministic equivalent of the primal resolvent, $(\hat{\Sigma} + \lambda I)^{-1} \mapsto (\Sigma + \kappa I)^{-1}$.) Clearing the denominator:

$$\kappa = \lambda + \kappa \sum_k \frac{\lambda_k}{P(\lambda_k + \kappa)}. \quad (7.15)$$

Ridgeless learning is $\lambda \rightarrow 0^+$ with κ remaining positive when P does not saturate every mode. Each mode contributes a factor $\lambda_k/(\lambda_k + \kappa)$ to n_{eff} , and κ is whatever is left of λ after that absorption.

In this equivalent, the coefficient of mode k concentrates on

$$\hat{a}_k = \frac{\lambda_k}{\lambda_k + \kappa} \bar{a}_k + \frac{\lambda_k}{\lambda_k + \kappa} \xi_k, \quad (7.16)$$

where ξ_k is mean-zero sampling noise of variance σ^2/P (the empirical inner product of ψ_k against the P noise samples).

7.5 Bias, variance, learning curve

Square-loss population risk is $R = \mathbb{E}_x[(\hat{f}(x) - f^*(x))^2] + \sigma^2$. Insert (7.16):

$$R = \sum_k t_k \left(\frac{\kappa}{\lambda_k + \kappa} \right)^2 + \frac{\sigma^2}{P} \sum_k \left(\frac{\lambda_k}{\lambda_k + \kappa} \right)^2 + \sigma^2. \quad (7.17)$$

The first sum is bias: the filter $\kappa/(\lambda_k + \kappa)$ leaves mode k unlearned until $P\lambda_k \gtrsim \kappa$. The second is variance from fitting noise.

If λ_k decays, as it does for NTK, Laplace, and ReLU kernels on the sphere, small- k modes reach $\lambda_k \gg \kappa$ first. The residual of mode k falls only once $P \gtrsim \kappa/\lambda_k$: *spectral bias*. For fixed $\{\lambda_k\}$, a target with t_k on large- λ_k modes has smaller R at given P : *task-model alignment*. As $\lambda \rightarrow 0$ and P crosses the rank of $\hat{K}$, $n_{\text{eff}} \rightarrow P$ forces κ through a peak and the variance term spikes (*double descent*); a decaying spectrum or $\lambda > 0$ smooths the spike.

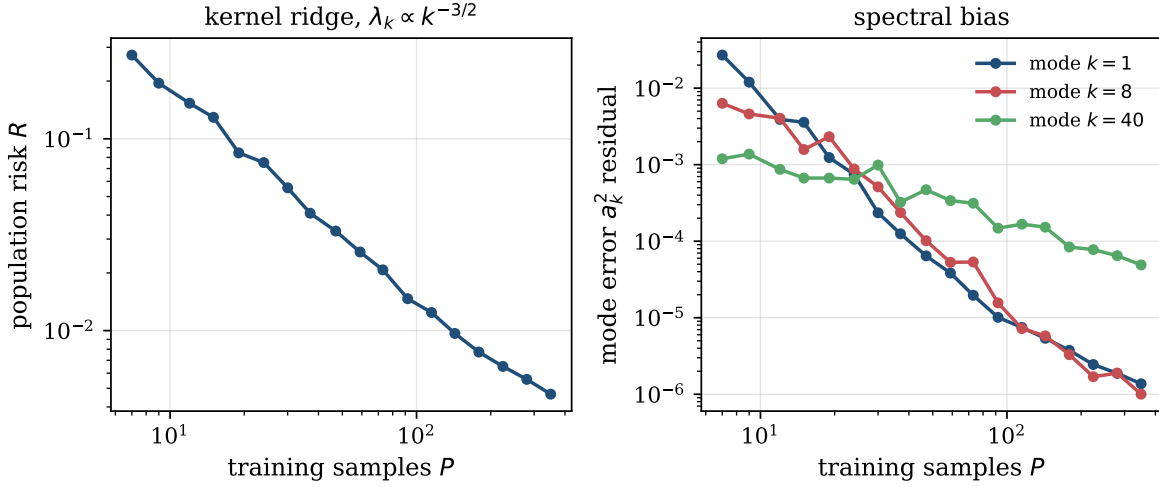


Figure 7: Kernel ridge with $\lambda = 10^{-3}$ and $\lambda_k \propto k^{-3/2}$. Left: population risk versus P . Right: residual power in modes $k = 1, 8, 40$. Averages over Gaussian designs.

Lazy neural generalization is (7.17) for the NTK spectrum $\Theta^{(L)}(t = 0)$. On the sphere that spectrum is a Gegenbauer transform of $\Theta^{(L)}$ [1].

Exercise 7.1. From (7.15), with $\sigma = 0$ and $\lambda_k = k^{-\beta}$, $\beta > 1$, estimate the P at which mode k is half-learned ($\kappa \approx \lambda_k$).

Solution. Half-learning of mode k means $\kappa \approx \lambda_k = k^{-\beta}$. Take $\lambda \rightarrow 0^+$ in (7.15):

$$1 = \sum_{j \geq 1} \frac{\lambda_j}{P(\lambda_j + \kappa)} = \frac{1}{P} \sum_{j \geq 1} \frac{1}{1 + \kappa j^\beta}.$$

Replace the sum by an integral and set $m = \kappa^{-1/\beta}$, so $\lambda_m = \kappa$ and $m \approx k$:

$$\sum_{j \geq 1} \frac{1}{1 + \kappa j^\beta} \approx \int_0^\infty \frac{dj}{1 + (j/m)^\beta} = m \int_0^\infty \frac{du}{1 + u^\beta}.$$

The u -integral converges at infinity iff $\beta > 1$, and is an $O(1)$ constant C_β . Thus $1 \approx C_\beta m/P$, i.e. $P \sim m \sim k$. Mode k is half-learned when the sample size reaches the mode index.

Answer. $P \sim k$. Equivalently $P \sim \kappa^{-1/\beta}$ with $\kappa = \lambda_k = k^{-\beta}$.

8. Effective theory at initialization

The infinite-width kernels still depend on depth through a nonlinear map. That map can explode, vanish, or sit at a fixed point, according to the initialization variances [2]. Finite-width corrections around the fixed point are organized in L/N .

8.1 Gaussian integrals

If $z \sim \mathcal{N}(0, K)$ in $\mathbb{R}^m$,

$$\mathbb{E}[e^{j \cdot z}] = \exp(\tfrac{1}{2} j^\top K j). \quad (8.1)$$

Odd moments vanish, and

$$\mathbb{E}[z_a z_b] = K_{ab}, \quad \mathbb{E}[z_a z_b z_c z_d] = K_{ab} K_{cd} + K_{ac} K_{bd} + K_{ad} K_{bc}. \quad (8.2)$$

For a general activation,

$$\mathbb{E}[\varphi(h)\varphi(h')] = \int \frac{du dv}{2\pi\sqrt{\det C}} \varphi(u)\varphi(v) \exp(-\tfrac{1}{2}(u, v) C^{-1}(u, v)^\top), \quad (8.3)$$

with $C = \begin{pmatrix} K_{xx} & K_{xx'} \\ K_{xx'} & K_{x'x'} \end{pmatrix}$.

8.2 Kernel recursion

Biases $b_i \sim \mathcal{N}(0, C_b)$ and rescaled weight variance C_W give

$$K_{\alpha\beta}^{(\ell+1)} = C_b + C_W \mathbb{E}[\varphi(z_\alpha)\varphi(z_\beta)]_{K^{(\ell)}}, \quad K_{\alpha\beta}^{(1)} = C_b + C_W D^{-1} x_\alpha \cdot x_\beta. \quad (8.4)$$

On the diagonal, $q_\ell \equiv K^{(\ell)}(x, x)$ is a 1d map $q \mapsto F(q)$. Off the diagonal one tracks $c_\ell = K^{(\ell)}(x, x')/\sqrt{q_\ell(x)q_\ell(x')}$.

Linearizing about a fixed point $q_\star = F(q_\star)$,

$$\chi \equiv F'(q_\star) = C_W \mathbb{E}[\dot{\varphi}(z)^2]_{z \sim \mathcal{N}(0, q_\star)}. \quad (8.5)$$

Then $\delta q_{\ell+1} = \chi \delta q_\ell + \dots$, and the off-diagonal perturbation δc is multiplied by the same χ . Hence $\chi < 1$ is ordered (signals vanish), $\chi > 1$ is chaotic (signals explode), and $\chi = 1$ is critical. The pair $(\chi = 1, q_\star = F(q_\star))$ fixes (C_W, C_b) for a given φ .

Deep linear. $\varphi(z) = z$ implies $K^{(\ell+1)} = C_W K^{(\ell)}$, so $K^{(\ell)} = C_W^{\ell-1} K^{(1)}$. Criticality is $C_W = 1$; any other choice explodes or vanishes exponentially in depth.

tanh. Odd, $\dot{\varphi}(0) = 1$. At $C_b = 0$ the origin $q_\star = 0$ is a fixed point and $\chi(0) = C_W$. Criticality at the origin is $C_W = 1$. For $C_W < 1$ the kernel collapses with depth; for $C_W > 1$ it grows until the nonlinearity saturates. The same pattern holds for sin and erf (the $K' = 0$ class of [2]).

ReLU. Homogeneous. The Cho–Saul formula (3.15) is the diagonal-plus-cosine map. On the diagonal, $z \sim \mathcal{N}(0, q)$ gives $\mathbb{E}[\varphi(z)^2] = q/2$, so $q_{\ell+1} = (C_W/2)q_\ell$ at $C_b = 0$. Fixed-point criticality is $C_W = 2$ (He initialization). Off the diagonal the cosine obeys the Cho–Saul angular map; there is no exponentially stable $q_\star > 0$ at $C_b = 0$ other than this ray.

8.3 Four-point functions and L/N

At finite N the empirical metric

$$G_{\alpha\beta} \equiv N^{-1} \sum_{i=1}^N h_i(x_\alpha) h_i(x_\beta) \quad (8.6)$$

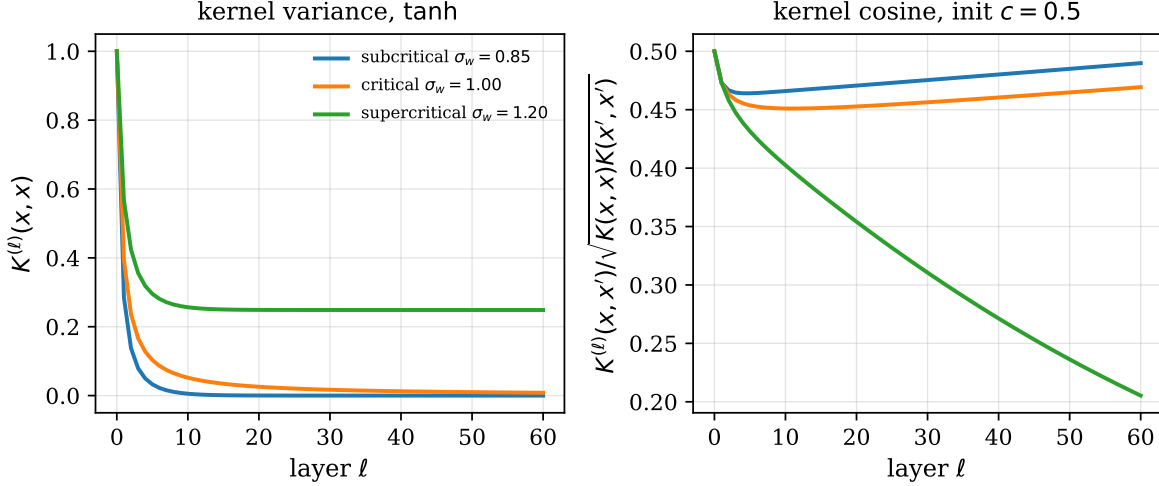


Figure 8: tanh kernel recursion (8.4) with $C_b = 0$. Left: $K^{(\ell)}(x, x)$. Right: cosine of two inputs, initialized at $c = 0.5$. Subcritical, critical ($C_W = 1$), and supercritical C_W .

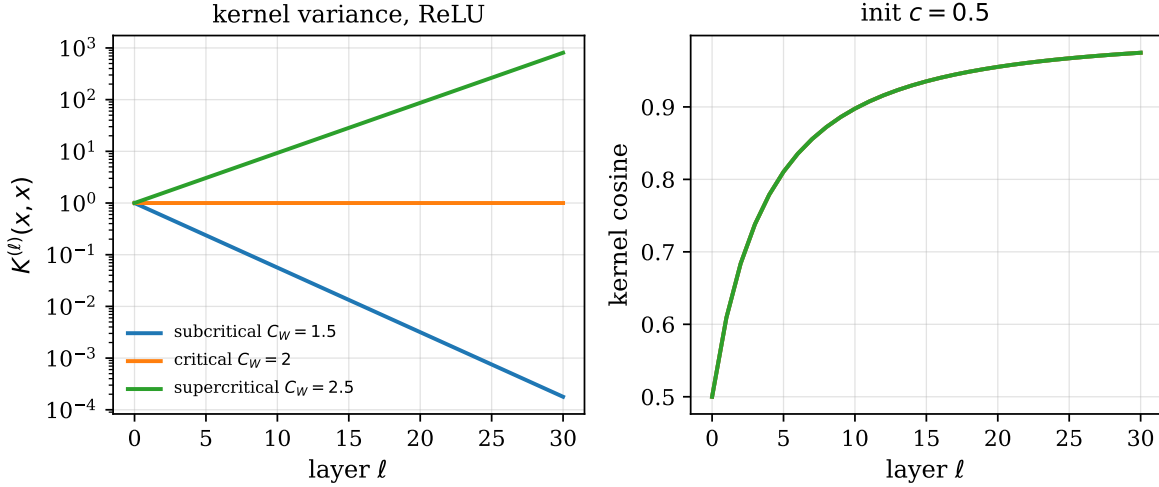


Figure 9: ReLU kernel recursion from (3.15), $C_b = 0$. Variance $q_{\ell+1} = (C_W/2)q_\ell$ is critical at $C_W = 2$.

fluctuates about $K_{\alpha\beta}$. For a *deep linear* net and a single input, $C_b = 0$, $C_W = 1$, the next-layer preactivations are $z = N^{-1/2}Wh$ with $W_{ij} \sim \mathcal{N}(0, 1)$ independent of h . Then $G' = N^{-1}\|z\|^2$ and, given h , each $z_i \sim \mathcal{N}(0, G)$ independently, so $\mathbb{E}[G' | h] = G$. Thus $\mathbb{E}[G'] = \mathbb{E}[G]$. For the second moment, Wick (8.2) on a pair of Gaussians gives $\mathbb{E}[z_i^4 | h] = 3G^2$ and $\mathbb{E}[z_i^2 z_j^2 | h] = G^2$ ($i \neq j$), hence

$$\mathbb{E}[G'^2 | h] = N^{-2}(N \cdot 3G^2 + N(N-1)G^2) = G^2(1 + \frac{2}{N}). \quad (8.7)$$

Taking $\mathbb{E}[\cdot]$ and subtracting $\mathbb{E}[G']^2 = K^2$,

$$\text{Var}(G') = (1 + \frac{2}{N})(\text{Var}(G) + K^2) - K^2 = \text{Var}(G) + \frac{2K^2}{N} + \frac{2}{N}\text{Var}(G). \quad (8.8)$$

At large N with $\text{Var}(G) = O(N^{-1})$ the last term is $O(N^{-2})$, and

$$\mathbb{E}[G] = K, \quad \text{Var}(G^{(\ell+1)}) = \text{Var}(G^{(\ell)}) + \frac{2K^2}{N}. \quad (8.9)$$

Iterating from a deterministic first layer,

$$\text{Var}(G^{(L)}) = \frac{2K^2L}{N}. \quad (8.10)$$

The same L/N appears for nonlinear φ at criticality, with a different $O(1)$ prefactor $V_0(\varphi)$ [2]. Infinite-width formulae are accurate when $L/N \ll 1$; $1/N$ corrections accumulate additively over depth.

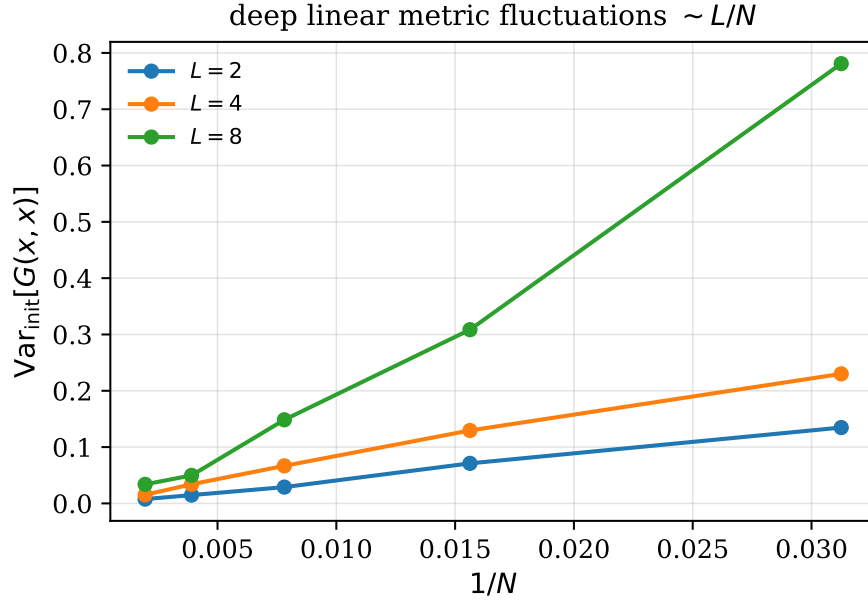


Figure 10: Deep linear MLPs at $C_W = 1$. Variance of $G(x, x) = N^{-1}\|h^{(L)}(x)\|^2$ across 400 initializations, versus $1/N$, for depths $L = 2, 4, 8$.

Exercise 8.1. Derive (8.9) from (8.2) for $z = N^{-1/2}Wh$ with $W_{ij} \sim \mathcal{N}(0, 1)$ independent of h .

Solution. Condition on h and write $G = N^{-1}\|h\|^2$. Rows of W are independent, so $z_i | h \sim \mathcal{N}(0, G)$ iid. Then $G' = N^{-1}\sum_i z_i^2$ has $\mathbb{E}[G' | h] = G$. Wick for a centred Gaussian is $\mathbb{E}[z_i^4 | h] = 3G^2$ and $\mathbb{E}[z_i^2 z_j^2 | h] = G^2$ for $i \neq j$, which is (8.7). Unconditioning produces (8.8); dropping the $O(N^{-2})$ term yields (8.9).

Answer. $\text{Var}(G') = \text{Var}(G) + 2K^2/N$ at leading order in $1/N$, since $\mathbb{E}[z^4] = 3G^2$ contributes the extra $2G^2/N$.

9. Finite-width representation learning

Infinite-width NTK dynamics are linear in function space: $f(t)$ stays in the affine space $f(0) + \text{span}\{K(x_\mu, \cdot)\}$. The first correction that moves the kernel is $O(L/N)$ at criticality [2] and $O(\gamma_0)$ at infinite width in μP .

9.1 Quadratic model

Taylor-expand in $\delta\theta = \theta - \theta_0$:

$$f(x; \theta_0 + \delta\theta) = f(x; \theta_0) + J(x) \cdot \delta\theta + \frac{1}{2} \delta\theta^\top H(x) \delta\theta + O(\|\delta\theta\|^3), \quad (9.1)$$

with $J(x) = \partial_\theta f(x)|_{\theta_0}$ and $H_{pq}(x) = \partial_{\theta_p} \partial_{\theta_q} f(x)|_{\theta_0}$. The first two terms are a linear model with features J ; their Gram matrix is the NTK

$$\Theta(x, x') = J(x) \cdot J(x'). \quad (9.2)$$

Truncating at linear order in $\delta\theta$ recovers frozen-NTK training. The quadratic term is the leading representation-learning correction.

9.2 Differential of the NTK

Along a parameter direction u ,

$$(\nabla_u \Theta)(x, x') = H(x) u \cdot J(x') + J(x) \cdot H(x') u. \quad (9.3)$$

The *trilinear* kernel

$$D(x, x', x'') \equiv \partial_\theta \Theta(x, x') \cdot \partial_\theta f(x'') = J(x'')^\top H(x) J(x') + J(x) \cdot H(x') J(x'') \quad (9.4)$$

is the dNTK [2]. It vanishes for a linear model ($H = 0$).

Chain rule on Θ along gradient flow (2.6), using $\Delta_\mu = y_\mu - f_\mu$ and $\nabla_\theta \mathcal{L} = -P^{-1} \sum_\mu \Delta_\mu J(x_\mu)$,

$$\partial_t \Theta(x, x') = \partial_\theta \Theta(x, x') \cdot \partial_t \theta = \frac{\eta}{P} \sum_\mu D(x, x', x_\mu) \Delta_\mu. \quad (9.5)$$

One GD step of size η is the same statement at $O(\eta)$:

$$\Delta \Theta(x, x') = \frac{\eta}{P} \sum_\mu D(x, x', x_\mu) \Delta_\mu + O(\eta^2). \quad (9.6)$$

In NTK scaling $\delta\theta = O(N^{-1/2})$, so D does not move Θ by an $O(1)$ amount. The first layer of an MLP contributes $D^{(1)} = 0$ (the map $x \mapsto W^{(1)}x$ is linear in $W^{(1)}$). Each subsequent layer sources a nonzero D from $\ddot{\varphi}$ and from products of earlier Θ 's. After L layers at criticality, $D = O(L)$, hence

$$\frac{\Delta \Theta}{\Theta} = O\left(\frac{L}{N}\right) \quad (9.7)$$

for NTK-scaled GD with $\eta = O(1)$. The same ratio is $O(1)$ in μP at infinite N .

9.3 Nearly-kernel prediction

Keep the quadratic term in (9.1) and take a stationary point of square loss at which the linear problem interpolates. The linear (NTK) solution is

$$\delta\theta_0 = \sum_{\mu=1}^P \alpha_\mu J(x_\mu), \quad \alpha = \hat{\Theta}^{-1}(y - F(0)), \quad (9.8)$$

which is (4.9) in parameter space. The quadratic correction at a test point is then

$$\frac{1}{2} \delta \theta_0^\top H(x) \delta \theta_0 = \frac{1}{2} \sum_{\mu\nu} \alpha_\mu \alpha_\nu J(x_\mu)^\top H(x) J(x_\nu). \quad (9.9)$$

The first summand of (9.4) is exactly $J(x_\mu)^\top H(x) J(x_\nu)$, so

$$f(x; \infty) = f_{\text{NTK}}(x) + \frac{1}{2} \sum_{\mu\nu} \alpha_\mu \alpha_\nu (D(x, x_\mu, x_\nu) - J(x) \cdot H(x_\mu) J(x_\nu)) + O(\|\delta \theta\|^3). \quad (9.10)$$

Predictions are still determined by a finite collection of kernels (Θ, D) together with the Hessian contracted against the NTK coefficients. The omitted $J(x) \cdot H(x_\mu) J(x_\nu)$ piece is the variation of the training Jacobians; it is the same order as D .

The kernel picture is accurate when $L/N \ll 1$ and $d < \frac{1}{2}$. Representation learning is $O(1)$ when $L/N = O(1)$, or when $d = \frac{1}{2}$ with $\gamma_0 = O(1)$. At fixed compute the useful aspect ratio is an $O(1)$ value of L/N : large enough that D can move features, small enough that the L/N expansion remains under control.

Exercise 9.1. For $f(x) = \frac{1}{2} \theta^2 x^2$ with scalar x and parameter θ , compute Θ and D , and verify (9.6) for one GD step.

Solution. Jacobians: $J(x) = \partial_\theta f = \theta x^2$ and $H(x) = \partial_\theta^2 f = x^2$. The NTK is

$$\Theta(x, x') = J(x) J(x') = \theta^2 x^2 x'^2.$$

The dNTK is

$$D(x, x', x'') = \partial_\theta \Theta(x, x') J(x'') = (2\theta x^2 x'^2)(\theta x''^2) = 2\theta^2 x^2 x'^2 x''^2.$$

One GD step on (2.5) updates $\theta \leftarrow \theta - \eta P^{-1} \sum_\mu (f_\mu - y_\mu) J_\mu$, i.e.

$$\Delta \theta = \frac{\eta}{P} \sum_\mu \Delta_\mu (\theta x_\mu^2), \quad \Delta_\mu = y_\mu - f_\mu.$$

The NTK changes as $\Delta \Theta(x, x') = ((\theta + \Delta \theta)^2 - \theta^2) x^2 x'^2 = (2\theta \Delta \theta + (\Delta \theta)^2) x^2 x'^2$. The $O(\eta)$ piece is

$$2\theta \Delta \theta x^2 x'^2 = \frac{\eta}{P} \sum_\mu (2\theta^2 x^2 x'^2 x_\mu^2) \Delta_\mu = \frac{\eta}{P} \sum_\mu D(x, x', x_\mu) \Delta_\mu,$$

which is (9.6). (The linear model $f = \frac{1}{2} A x^2$ in the parameter A has $H = 0$ and $D = 0$; the θ^2 parameterization is the minimal example with a nonzero dNTK.)

Answer. $\Theta = \theta^2 x^2 x'^2$, $D = 2\theta^2 x^2 x'^2 x''^2$. One GD step reproduces (9.6) at $O(\eta)$.

A. Definitions

Gram matrix.

For vectors $v_1, \dots, v_P$, the matrix $G_{\mu\nu} = v_\mu \cdot v_\nu$. Always positive semidefinite. The kernel matrix $\hat{K}_{\mu\nu} = K(x_\mu, x_\nu)$ is the Gram matrix of the feature maps $\Phi(x_\mu)$.

Positive-definite kernel.

A symmetric $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ such that $\sum_{\mu\nu} c_\mu c_\nu K(x_\mu, x_\nu) \geq 0$ for all finite collections $\{x_\mu\}$ and all $c \in \mathbb{R}^P$. Equivalent, by Mercer, to $K(x, x') = \Phi(x) \cdot \Phi(x')$ for some (possibly infinite-dimensional) feature map Φ .

Feature map / kernel trick.

A lift $\Phi : \mathcal{X} \rightarrow \mathcal{H}$ into a Hilbert space. Inner products in $\mathcal{H}$ can be evaluated as $K(x, x')$ without writing Φ down. The SVM dual (3.5) and kernel ridge (3.9) depend on data only through K .

RKHS.

The reproducing kernel Hilbert space $\mathcal{H}_K$ of a kernel K is the completion of $\text{span}\{K(x, \cdot) : x \in \mathcal{X}\}$ under the inner product $\langle K(x, \cdot), K(x', \cdot) \rangle_{\mathcal{H}_K} = K(x, x')$. Evaluation is continuous: $h(x) = \langle h, K(x, \cdot) \rangle_{\mathcal{H}_K}$.

Representer theorem.

Empirical-risk minimizers in $\mathcal{H}_K$ with an RKHS-norm penalty lie in $\text{span}\{K(x_\mu, \cdot)\}_{\mu=1}^P$. The infinite-dimensional problem collapses to P coefficients.

Support vector.

A training point with nonzero dual coefficient α_μ in (3.4). Complementary slackness puts it on the margin. The max-margin hyperplane is a linear combination of support vectors only.

Cho–Saul formula.

The kernel of a random ReLU feature $\varphi(z) = \max(z, 0)$. For $w \sim \mathcal{N}(0, D^{-1}I)$ and $\vartheta = \arccos(x \cdot x' / (\|x\| \|x'\|))$,

$$\mathbb{E}[\varphi(w \cdot x) \varphi(w \cdot x')] = \frac{\|x\| \|x'\|}{2\pi D} (\sin \vartheta + (\pi - \vartheta) \cos \vartheta).$$

This is (3.15). The NNGP recursion for deep ReLU nets is the iteration of this map.

Gradient flow.

The ODE $\dot{\theta} = -\eta \nabla_{\theta} \mathcal{L}(\theta)$, the $\eta \rightarrow 0$ limit of gradient descent. In these notes, the source of all training dynamics.

Gaussian process.

A random function f such that $(f(x_1), \dots, f(x_n))$ is jointly Gaussian for every finite set. Specified by a mean and a covariance kernel $K(x, x') = \mathbb{E}[f(x)f(x')]$. An infinite-width net at initialization is a GP with the NNGP kernel.

NNGP.

Neural network Gaussian process: the $N \rightarrow \infty$ law of $f(\cdot; \theta)$ at initialization. Covariance given by the feature-kernel recursion (6.2).

Neural tangent kernel.

$K(x, x'; t) = \partial_{\theta} f(x; t) \cdot \partial_{\theta} f(x'; t)$. The metric that makes gradient flow on θ into an ODE on functions (4.3). Frozen if features do not move.

Trilinear kernel.

A scalar $T(x, x', x'')$ linear in each slot. The dNTK $D(x, x', x'') = \partial_{\theta} \Theta(x, x') \cdot \partial_{\theta} f(x'')$ is trilinear in the Jacobians and Hessians of f : it contracts $H(x)$, $J(x')$, and $J(x'')$.

Order parameter.

A macroscopic summary (here Φ , G , K , Θ) that concentrates as $N \rightarrow \infty$. The microscopic weights fluctuate; the order parameters obey a closed equation.

NTK parameterization / $\mu\mathbf{P}$.

Two scalings of (5.1) with width. NTK parameterization: $d = 0$, features freeze, $K(t) = K(0)$. Maximal update parameterization ($\mu\mathbf{P}$) / mean-field: $d = \frac{1}{2}$, features move by $O_N(1)$, kernels obey DMFT. See (5.11).

DMFT.

Dynamical mean-field theory. After $N \rightarrow \infty$, each neuron is an independent copy of a stochastic process whose statistics (Φ, G) are determined self-consistently from that same process. Replaces $O(N)$ coupled ODEs by $O(P^2)$ kernel equations.

Criticality.

Initialization (C_W, C_b) at which the kernel recursion has $\chi = F'(q_\star) = 1$. Signals neither explode nor vanish with depth. For tanh at $C_b = 0$, $C_W = 1$; for ReLU, $C_W = 2$.

Stability parameter χ .

$\chi = C_W \mathbb{E}[\dot{\varphi}(z)^2]$ at the kernel fixed point. Multiplies perturbations of K from layer to layer. $\chi = 1$ is critical.

Depth-to-width ratio.

L/N . Leading finite-width expansion parameter at criticality: $\text{Var}(G) \sim (L/N)V_0$. Kernel formulae hold for $L/N \ll 1$; dNTK representation learning is $O(L/N)$.

dNTK.

Differential of the NTK, $D(x, x', x'') = \partial_\theta \Theta(x, x') \cdot \partial_\theta f(x'')$. Trilinear kernel that sources $\dot{\Theta}$ at finite width. Vanishes for linear models and, to leading order, in infinite-width NTK scaling.

Wick's theorem.

Gaussian moment factorization: $\mathbb{E}[z_{a_1} \cdots z_{a_{2n}}]$ is the sum over pairings of products of covariances. Used to close four-point recursions for deep linear nets, (8.9).

Deterministic equivalence.

In high dimension, a random matrix functional (here the ridge resolvent $(\hat{\Sigma} + \lambda I)^{-1}$) concentrates on a non-random proxy, typically involving an effective ridge κ . Converts the random learning curve into (7.17).

Spectral bias.

Kernel machines learn eigenmodes of K in decreasing order of λ_k . Mode k is fit once $P\lambda_k \gtrsim \kappa$.

Lazy / rich training.

Lazy: $K(t) \approx K(0)$, the net is a linear model in its initial Jacobian features. Rich: hidden features $\Phi(t)$ move by $O_N(1)$. Selected by d in (5.2), or by γ_0 at finite N .

References

- [1] Cengiz Pehlevan and Blake Bordelon. Lecture notes on infinite-width limits of neural networks. Princeton ML Theory Summer School, 2023.

- [2] Daniel A. Roberts, Sho Yaida, and Boris Hanin. *The Principles of Deep Learning Theory*. Cambridge University Press, 2022. <https://deeplearningtheory.com>.